Report on Implemented Programs (Haoran Ma)

Multiset Implementation

The Multiset class is implemented as a balanced binary search tree, ensuring efficient insertion, deletion, and lookups. It manages duplicate values by maintaining a count for each key, allowing multiple occurrences of the same element. The class also provides functions like Min, Max, Floor, and Ceil for precise element retrieval, enhancing its usability in various scenarios.

To verify correctness, the Google Test framework was expanded to include:

- Duplicate handling: The MultipleDuplicates test ensures that inserting the same element multiple times properly updates its count and reflects the correct total size.
- Removal tests: The RemoveDuplicate test checks that sequential removals decrement the count and eventually remove the element, while RemoveNonexistent ensures an exception is thrown when trying to remove a missing element.
- Lookup validation: Tests for Contains, Min, Max, Floor, and Ceil confirm correct behavior, including handling boundary conditions.

Prime Factorization Program

The prime_factors.cc program performs prime factorization using the Multiset class to store and count the occurrences of prime factors. The program follows a trial division approach, iterating through possible divisors and inserting discovered prime factors into the multiset. If the only factor found is the number itself, it is classified as a prime.

The program supports several command-line options:

- all - Displays all prime factors along with their frequencies.
- min/max - Retrieves the smallest and largest prime factors, respectively.
- near - Finds the closest prime factor relative to a given value using Floor and Ceil operations.